Plants in limbo – the theory of detrital photosynthesis

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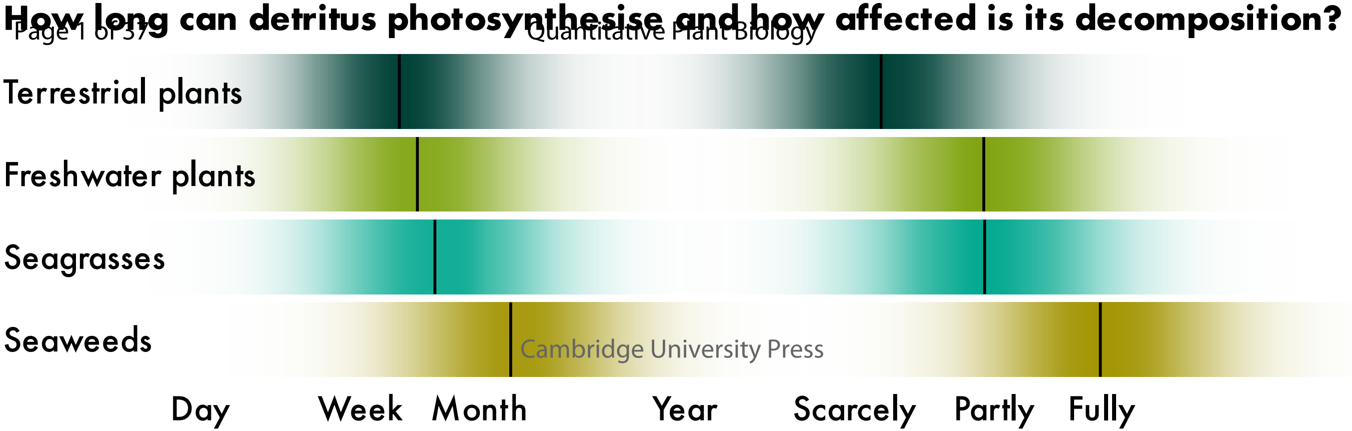
How long can detritus photosynthesise and how affected is its decomposition?
Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Day
Week
Month
Year
Scarcely
Partly
Fully

Quantitative Plant Biology

Theories

Plants in limbo — the theory of detrital photosynthesis

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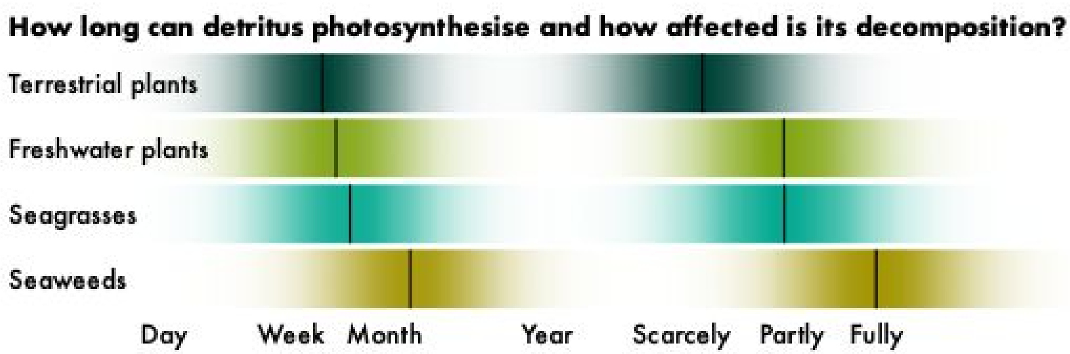
Abstract



Kelp detritus can maintain photosynthesis following detachment. This could be explained by limited tissue differentiation in non-vascular plants, but it remains unclear if detrital photosynthesis is restricted to macroalgae. Here I formalise the theory of detrital photosynthesis by developing mathematical and causal models and performing a meta-analysis of post-detachment physiology across 129 studies on 92 species of terrestrial and aquatic plants, including original data on the seagrasses *Amphibolis antarctica* and *Halophila ovalis*. Detrital photosynthesis is a general phenomenon, but its longevity in macroalgae is unmatched. Median photosynthetic half-life increases from terrestrial plants (11 days) over freshwater Alismatales (14 days) and seagrasses (17 days) to seaweeds (44 days) and only matches or exceeds detrital half-life for seaweeds (3:1). Detrital photosynthesis therefore uniquely determines macroalgal as opposed to vascular plant decomposition. This explains why light slows kelp decomposition, contrary to the expected effect of photodegradation. The presented theory questions conventional understanding of decomposition and the carbon cycle.

Keywords decay, degradation, leaf litter, biogeochemistry, blue carbon

Introduction

…tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end… — Philip Pullman (1994) *The Tin Princess*

Seaweed detrital dynamics are influenced by detrital photosynthesis. Kelp detritus can remain physiologically viable for months (de Bettignies et al. 2020, Frontier et al. 2021, Wright & Foggo 2021, Frontier et al. 2022, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024), often resulting in atypical decomposition trajectories (Frontier et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Kennedy & Blain 2025). Floating kelp detritus sinks before it stops photosynthesising (Graiff et al. 2013, 2016, Tala et al. 2019). And if detached tissue happens to be reproductive, it can even be a vector for dispersal (Macaya et al. 2005, Fraser et al. 2018, Tala et al. 2019, de Bettignies et al. 2020, Frontier et al. 2021). Detrital photosynthesis — defined as the persistence of photosynthesis after detachment (Frontier et al. 2021, Wright et al. 2022, 2024) — provides the strongest evidence in support of the viability of detritus. Its fundamental impact on detrital dynamics only becomes apparent when comparing decomposition rates between live and dead or senescent detritus: seaweed detritus with absent or impaired physiology clearly decomposes faster (Birch et al. 1983, Brouwer 1996, Hamersley et al. 2015, de Bettignies et al. 2020). The effect of detrital photosynthesis on decomposition is likely due to maintenance of growth, repair and induced chemical defence (Tala et al. 2019, de Bettignies et al. 2020, Wright et al. 2022), probably making physiology an important source of detrital recalcitrance (Wright et al. 2022, 2024). This notion stands in stark contrast to the established ecological definition of detritus as "dead organic matter" (Moore et al. 2004, discussed in Wright & Kregting 2023). One solution to this conundrum is to insist that detached plant material is not detritus provided that it is still alive. However, detached plants (except for holopelagic and pleustophytic species) are commonly referred to as detritus. Moreover, detachment clearly demarcates the detrital phase but the limbo following detachment remains vague. I therefore continue to use the term "detritus" for any detached plant tissue and "detritus sensu stricto" for detached *and* dead plant tissue.

There is currently no reference point for detrital photosynthesis. The term was first introduced quite recently in the context of kelp decomposition (Frontier et al. 2021, Wright et al. 2022, 2024). But there was clearly awareness of the phenomenon among early phycologists who noted the ability of excised tissue to maintain photosynthesis (Drew 1983) and deemed it necessary to pre-kill detritus to measure decomposition (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). The term "detrital photosynthesis" is not used in other fields of botany but the amount of research into physiology of excised leaves of terrestrial and freshwater plants is enormous (e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, 1982, Kar & Choudhuri 1986, 1987, Powles et al. 2006, Gauthier & Jacobs 2018, Kar et al. 2021). The literature also includes three studies on seagrasses (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986), the only fully marine angiosperms and vascular counterpart to seaweeds (Olsen et al. 2016). The aims of measuring photosynthesis in detached leaves are varied, with most studies using excision as an experimental proxy for plant senescence. No attempt has been made to align these seemingly incongruous data in a formal comparison across plant groups (sensu Bolton 2016). It consequently remains unclear if detrital photosynthesis is unique to seaweeds. Although it has been suggested that detrital photosynthesis may be explained by the ability of all macroalgal tissues to independently photosynthesise to some extent (Wright & Kregting 2023), there is no certainty due to a lack of knowledge on other plants. The mechanism underlying the ability of seaweeds to maintain detrital photosynthesis remains unclear. It could simply be their aquatic nature which removes reliance on roots for water and nutrient provisioning. Or it could be more deeply rooted in their non-vascularity which requires minimal tissue differentiation. To understand the prerequisites for detrital photosynthesis and thus predict prevalence among plants, model systems beyond seaweeds are needed.

Seagrasses are suitable candidates for further investigation of detrital photosynthesis. While they are also marine macrophytes, seagrasses are fundamentally different from seaweeds. Seagrasses are vascular plants which, due to their terrestrial ancestry (Olsen et al. 2016), are differentiated into roots, rhizomes or stems, and leaves. However, convergent evolution with seaweeds (Olsen et al. 2016) has reduced their reliance on roots for nutrient uptake. Seagrass leaves tend to outperform their roots in suppling nutrients to the plant (Pedersen & Borum 1992, Stapel et al.

1996, Pedersen et al. 1997, Gras et al. 2003, Viana et al. 2019). This suggests that seagrass leaves can independently photosynthesise, seemingly reducing the primary function of the roots to anchorage, analogous to the holdfast of seaweeds. However, roots can supply more nitrogen despite the leaves' greater uptake affinity because of the higher ammonium concentration in sediment pore water (Short & McRoy 1984, Lee & Dunton 1999, but see Pedersen & Borum 1992) and tap into nutrient pools that are unavailable to leaves, such as particulate (Evrard et al. 2005) and microbially fixed (Patriquin & Knowles 1972, Mohr et al. 2021) nitrogen. Nutrients are also translocated from roots to leaves (Short & McRoy 1984, Viana et al. 2019), even those tens of centimetres from the source (Marbà et al. 2002). Clearly the photosynthetic emancipation of seagrass leaves is at a halfway point between terrestrial plants and macroalgae, so seagrasses probably display detrital photosynthesis to some extent. Relatively slow and light-dependent chlorophyll degradation in seagrass detritus (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986) provides some support this. Seagrass detritus also often incudes green leaves (Knauer & Ayers 1977, Ochieng & Erftemeijer 1999, Jiménez-Ramos et al. 2023), likely owing to low nutrient resorption prior to abscission (Harrison 1989, Pedersen & Borum 1992, Pedersen et al. 1997, Hemminga et al. 1999, Rosch & Koch 2009) and premature detachment (Cebrián et al. 1997, Jiménez-Ramos et al. 2023). However, seagrass detrital photosynthesis remains to be directly investigated.

Here I aim to formalise the theory of detrital photosynthesis. I hope that this will provide a better conceptual foundation for the investigation of plant decomposition and carbon cycling more generally. To address this aim, I approached detrital photosynthesis holistically for the first time. I conducted a brief ex situ decomposition experiment on seagrasses to fill the knowledge gap on seagrass detrital photosynthesis. I chose the paddle weed *Halophila ovalis* (Hydrocharitaceae) and wire weed *Amphibolis antarctica* (Cymodoceaceae) as diverse representatives because of their phylogenetic (diverged 105 Ma ago, Waycott et al. 2018) and morphological (de los Santos et al. 2012, 2016) differences. Senescence of fresh leaves was induced by cutting their petiole and leaving them to decompose on sediment, measuring net oxygen production of destructively sampled leaves at regular intervals over 34 days. On the basis of these and previous (e.g. Wright et al. 2022, 2024) observations of detrital photosynthesis, I then developed a mathematical model which describes detrital photosynthesis as a logistic decay function of detrital age. After compiling a detrital photosynthesis dataset (Wright 2025), I used this model in a Bayesian meta-analysis to infer the longevity of detrital photosynthesis for different plants (sensu Bolton 2016) and quantify the effects of light and choice of response variable (photosynthesis vs. chlorophyll). I compared the persistence of detrital photosynthesis with detrital half-life (sensu Olson 1963) to determine relative influence of physiology on decomposition for different plants. Finally, I present a new causal model of decomposition that emerges naturally from the theory of detrital photosynthesis.

Methods

Model

Detached plant parts can remain in limbo for long durations, often photosynthesising at similar rates to the attached plant. For this reason and for simplicity, detrital photosynthesis has mostly been described as a linear function of detrital age (Frontier et al. 2021, Wright & Foggo 2021, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024). But in two cases a logistic decay function was used: in a beta model predicting chlorophyll fluorescence (Fv/Fm) (Frontier et al. 2022) and in a binomial model predicting the probability of photosynthesis (Wright et al. 2024) with time after excision. The logistic function describes the decay of photosynthesis after detachment better than the linear function because it is bounded. Detrital photosynthesis cannot conceivably exceed baseline photosynthesis under identical conditions. An ill-defined baseline or changes in environment (e.g. when detritus floats and is exposed to more light) may naturally lead to brief increases after detachment (Frontier et al. 2021, Wright & Kregting 2023). However, this cannot be considered part of the underlying photosynthetic decay process of interest. Similarly, photosynthesis cannot decline beyond a set minimum, which is zero for most response variables (e.g. gross photosynthesis, chlorophyll fluorescence and chlorophyll) and a negative constant for net photosynthesis. The logistic function is therefore the natural choice.

There are various ways to parameterise the logistic function. Typically, a simple linear term with intercept and slope is inserted into the standard logistic function. But for logistic decay with time (t), the intercept is not a meaningful parameter because $f(t = 0)$ is the known baseline. The slope is more meaningful because it provides the logistic rate of decay per unit t, but ultimately the parameter of interest is the sigmoid inflection point or midpoint: t when $f(t) = 0.5$. It has the same units as t and is the only parameter that gives a directly interpretable measure of decay. As such it is analogous to half-life ($t_{½}$), t when $f(t) = 0.5$ for the exponential decay function (Olson 1963). The logistic midpoint can be calculated as intercept divided by slope, but this usually results in enormous uncertainty and difficulty to calculate meaningful central tendencies (Wright et al. 2024), so the midpoint is best directly estimated. The slope and midpoint parameterisation is sometimes used but these parameters are not identifiable with an intercept near a known baseline. To simplify the model while retaining the midpoint, one must either fix the slope or intercept. Only a constant intercept is sensible and since the baseline is expected near 1, a logistic intercept of 5, translating to $f(0) = 0.99$, is the natural choice.

The most complex case to model is that of net photosynthesis since it has a negative lower bound. Two scaling parameters outside the standard logistic function are required. Thus my full logistic function describing the decay of photosynthesis (p) with time after detachment (t) is

$$p(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t-\mu)}{\mu}}} - \tau \tag{1}$$

where α is baseline photosynthesis, μ is the midpoint of photosynthetic decay and τ is baseline respiration. $\tau = 0$ when the lower bound is zero as for gross photosynthesis etc. and $\alpha = 1$ when photosynthesis is normalised relative to the baseline. The naming of parameters is inspired by the word אמת (graecised τμα) which is composed of the first, middle and last letters of the Hebrew alphabet, symbolising life (א), transition (מ) and death (ת). In the story of the golem (McElreath 2020), the word lends life but removing א changes its meaning from truth to death (מת) and takes life.

Literature review

Following the seagrass experiment (see *Supplementary methods*), I searched the literature using keyword strings such as "(photosynthesis OR chlorophyll) AND ('induced senescence' OR detrit* OR detach* OR excis* OR cut OR isolate*) AND (week* OR day* OR hour* OR minute* OR time OR age)" in Web of Science (webofscience.com/wos) and Google Scholar (scholar.google.com). Once I had a selection of suitable papers, I used Crossref Metadata Search (search.crossref.org) to find similar papers. I accepted studies on all plants (sensu Bolton 2016), the condition being that senescence was induced by excision at an exact time and photosynthesis was repeatedly measured in a timeseries of at least 3 timepoints over any interval to assess viability of the detached tissue. Studies on abscission were not considered since senescence starts at an unknown time prior to detachment. I naturally did not include plants that always live detached, such as pleustophytes and the holopelagic seaweeds *Sargassum fluitans* and *S. natans*, for which detrital age cannot be defined as time since detachment. I accepted various response variables, including O_2 and CO_2 gas exchange, ^{14}C assimilation, Fv/Fm, photosystem I and II electron transport rates (Mehler and Hill reactions), RuBisCo activity or concentration, carbonic anhydrase activity and photorespiration (glycolate concentration), as well as total chlorophyll, chlorophyll *a* or a chlorophyll indicator. Microbial photosynthesis was generally assumed to be negligible since most of the mentioned response variables would likely not detect it. Including the present study, this resulted in 129 suitable studies between 1957 and 2025 (see *Supplementary references*) and 551 experiments (median = 3 per study).

Data were extracted by copying tables or digitising figures using WebPlotDigitizer v4.6 (Rohatgi 2022). In two cases I contacted authors for their raw data and in five cases the data were my own. Raw data were otherwise not available. The resulting dataset (Wright 2025) contains data on 92 species from 23 orders in four non-taxonomic groups of interest: terrestrial plants (66 species), freshwater plants (6 species), seagrasses (5 species) and seaweeds (15 species). It is noteworthy that all freshwater plants in the dataset belong to the order Alismatales, like seagrasses, and all seaweeds belonged to the phyla Chlorophyta and Heterokontophyta (Table S1). The assigned taxonomy follows Plants of the World Online (powo.science.kew.org , (POWO 2025) and AlgaeBase (algaebase.org, Guiry, M.D. & Guiry 2025). My distinction between freshwater and terrestrial plants is based on whether leaves are submerged and photosynthesis can therefore be considered aquatic. There were no studies on freshwater macroalgae, so seaweeds are assumed to be representative. Several predictor variables including light- and water availability as well as the type of photosynthesis measurement were also recorded. When individual observations could not be extracted, the mean, standard error of the mean (s.e.m.) and sample size (n) were recorded instead. Uncertainty was mostly given as s.e.m., but when standard deviation (sd) was provided instead, it was converted as $\mathrm{sem} = \frac{\mathrm{sd}}{\sqrt{\mathrm{n}}}$. In a few cases uncertainty was given as a 95% confidence interval (CI) or interquartile range (IQR). In the former case, I converted as sem = $\frac{0.5 \times CI_{0.95}}{\Phi^{-1}(0.975)}$, in the latter I assumed mean = median and conservatively converted the larger of the two quartile ranges ($Q_3 - Q_2$ or $Q_2 - Q_1$) as sem = $\frac{QR}{\Phi^{-1}(0.75) \times \sqrt{n}}$.

Meta-analysis

As is typical of meta-analyses, the data contain a mixture of observations and means. To jointly analyse data with at different levels of uncertainty one must either summarise observations and build a measurement error model or expand means to observations by simulation. I opted for the latter and obtained 10646 observations by simulating draws from the sampling distribution. For most response variables, which are bounded by zero, I chose the gamma distribution, for net photosynthesis I chose the normal distribution and for Fv/Fm I chose a truncated normal distribution because some summary statistics violated the beta distribution. I implemented the sampling in R v4.5.2 (R Core Team 2026) using "rgamma(n , mean^2 / sd^2 , mean / sd^2)", "rnorm(n , mean , sd)" and "rtnorm(n , mean , sd , 0 , 1)", the last function being from extraDistr v1.10.0 (Wolodzko 2023).

The response variable is an amalgam of various measures of photosynthesis. I therefore normalised it to enable formal comparison. Observations were expressed relative to the initial observation or mean of initial observations in the timeseries. The resulting ratios, representing relative photosynthesis, are the least invasive form of normalisation and frequently used to express photosynthetic decay (29 of 129 studies, e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, Pellikaan 1982). The drawback of this approach is that data are not fully brought onto the same scale because net photosynthesis allows zeros and negative ratios. I resolved this issue by replacing all zeros and negative ratios with the minimum observed positive ratio, assumed to be the minimum observable ratio within measurement error of zero. This means equating net respiration with absence of photosynthesis, which is of course not the case because respiration can mask photosynthesis in net photosynthesis measurements. Nonetheless, I believe this to be acceptable for the purposes of determining the timescales of photosynthetic decay. This normalisation allows using the simplest form of Equation 1 with $\tau = 0$ and $\alpha = 1$.

Apart from t (days), the main predictor variables of interest are plant group (G = terrestrial plants, freshwater plants, seagrasses and seaweeds), light availability and the type of response variable (chlorophyll or photosynthesis). Light is naturally a key predictor of photosynthetic persistence (Frontier et al. 2021, 2022, Wright et al. 2024) and the type of response variable is important because chlorophyll and photosynthesis are expected to senesce on different timescales. Of the recorded predictors, water availability is only relevant to terrestrial plants and was therefore not considered in the model. Data coverage is incomplete across the main predictor variables, with data missing for the combinations freshwater–light–photosynthesis, seagrass–dark–photosynthesis and seaweed–dark–chlorophyll. Since light availability and the type of response variable are binary predictors, they were coded as indicators of light ($L = -0.5$ for dark, $+0.5$ for light) and chlorophyll ($C = -0.5$ for photosynthesis, $+0.5$ for chlorophyll), centred on zero to avoid the bias in prior variance typical of indicator variables (McElreath 2020) and allow easy prediction of the mean. To get a sense of interspecific and experimental variation, I also included plant species (S, 92 levels) and experiment (E, 551 levels) as predictors. The resulting Bayesian log-normal multilevel model of relative photosynthesis (p) in standard statistical notation (sensu McElreath 2020) is

$$\ln p_i \sim \mathrm{Normal}(\,\overline{p}_i\,,\sigma\,) \qquad (2)$$

$$\overline{p}_i = \ln \frac{1}{1 + e^{\frac{5(t_i - \mu_i)}{\mu_i}}}$$

$$\ln \mu_i = \alpha_{G_i} + \alpha_{S_i} + \alpha_{E_i} + \beta_{G_i} L_i + \gamma_{G_i} C_i$$

$$\alpha_G \sim \mathrm{Normal}(\,\overline{\alpha}\,, \sigma_{\alpha_G}\,)$$

$$\alpha_S \sim \mathrm{Normal}(\,0\,, \sigma_{\alpha_S}\,)$$

$$\alpha_E \sim \mathrm{Normal}(\,0\,, \sigma_{\alpha_E}\,)$$

$$\beta_G \sim \mathrm{Normal}(\,\overline{\beta}\,, \sigma_{\beta}\,)$$

$$\gamma_G \sim \mathrm{Normal}(\,\overline{\gamma}\,, \sigma_{\gamma}\,)$$

$$\overline{\alpha} \sim \mathrm{Normal}(\,\ln 14\,, 0.3\,)$$

$$\overline{\beta}, \overline{\gamma} \sim \mathrm{Normal}(\,0, 0.4\,)$$

$$\sigma_{\alpha_G}, \sigma_{\alpha_S}, \sigma_{\alpha_E} \sim \mathrm{Normal}_{0,\infty}(\,0, 0.3\,)$$

$$\sigma_{\beta}, \sigma_{\gamma} \sim \mathrm{Normal}_{0,\infty}(\,0, 0.4\,)$$

$$\sigma \sim \mathrm{Exponential}(\,1\,)$$

where the logarithm of μ (Equation 1) is described by a linear model with intercepts (α) indexed by G, S and E along with the effects of light (β) and measuring chlorophyll instead of photosynthesis (γ) indexed by G. β and γ are differences on the logarithmic scale, so represent log ratios. All parameters are partially pooled to enable prediction for new categories, borrow information and regularise spurious effects (McElreath 2020). The intercept standard deviations (σ_α) give an indication of intergroup, interspecific and experimental variance. The prior for the mean of μ on the logarithmic scale ($\overline{\alpha}$) is centred on ln 14 because two weeks seems like a reasonable median midpoint of photosynthetic decay based on my observations on kelps (Wright et al. 2022, 2024) and literature review. Even though I suspected β and γ to be positive, I centred the priors for $\overline{\beta}$ and $\overline{\gamma}$ on zero to avoid biasing the effect.

The model was written in Stan (Carpenter et al. 2017) and implemented with CmdStan v2.17.1 (Lee et al. 2026) in R. This was facilitated by cmdstanr v0.9.0 (Gabry et al. 2025), tidybayes v3.0.7 (Kay 2024) and bayesplot v1.14.0 (Gabry et al. 2019). Posterior probability distributions for each parameter were estimated based on 8×10^4 Markov chain Monte Carlo samples (10^4 iterations × 8 chains). Model performance was assessed using $\hat{R}$ scores and trace rank and pair plots (Gabry et al. 2019). For data wrangling, summary and visualisation I used tidyverse v2.0.0 (Wickham et al. 2019), here v1.0.2 (Müller 2025), ggridges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For details, see data and code at github.com/lukaseamus/plant-limbo. Vector graphics were polished in Affinity Designer v1.10.8 (Serif Ltd., Nottingham, UK).

To contextualise μ (days), I formally compared it with the half-life $t_{½}$ (days), its equivalent in the exponential model of decomposition. To facilitate the comparison, I hereinafter refer to μ as the photosynthetic half-life and $t_{½}$ as the detrital half-life. $t_{½}$ is related to the exponential decay constant k by $t_{½} = \ln 2 / k$ (Olson 1963). k is usually estimated using the exponential decay function but can also be approximated by the detrital turnover rate (P/B), the ratio of detrital production to the detrital pool (Cebrián & Lartigue 2004). Using published datasets (Cebrián & Lartigue 2004, Anderson-Teixeira et al. 2018, Filbee-Dexter et al. 2024, Kennedy & Blain 2025, White & Norkko 2025, Wright 2026a), I thus extracted or calculated k for terrestrial plants, freshwater plants, seagrasses and seaweeds, and converted it to $t_{½}$. I then modelled $t_{½}$ with a log-normal multilevel intercept model (see github.com/lukaseamus/plant-limbo for details) and compared its predictions to those for μ from Equation 2. I formalised this comparison as the ratio $\mu/t_{½}$, a measure of physiological influence on decomposition. $\mu/t_{½} = 1$ implies that detrital photosynthesis reaches half its initial value at the same time as detrital mass. If $\mu/t_{½} < 1$, photosynthesis decays faster than mass and is unlikely to influence decomposition. $\mu/t_{½} > 1$ is theoretically impossible because there can be no photosynthesis without mass, but in practice it will occur due to empirical uncertainty. $\mu/t_{½} > 1$ may be interpreted as strong indication that photosynthesis determines decomposition. $\mu/t_{½}$ was calculated for each plant group from the independently estimated posterior probability distributions for μ and $t_{½}$. This approach propagates all uncertainty and allows calculation of the probability that the ratio falls above or below 1.

Results

Seagrasses

Halophila ovalis photosynthesised for longer post-excision than *Amphibolis antarctica*. After 34 days, the duration of the experiment, all excised *H. ovalis* leaves were still obviously net autotrophic while all excised *A. antarctica* leaves were apparently dead. On a mass basis, *H. ovalis* displayed similar baseline light-saturated net photosynthesis (P_{max}) to *A. antarctica*, but its photosynthetic half-life (μ in Equation 1) was 3.9 ± 1.6 (mean ± standard deviation) times as long (Figure 1, Table 1). On a leaf basis, baseline P_{max} of *A. antarctica* was naturally much greater but again the photosynthetic half-life of *H. ovalis* was 6.8 ± 2.5 times as long (Table 1).

Mean leaf blotted mass of *H. ovalis* (initially 0.052 ± 0.005 g leaf^{-1}) declined at linear rates of 0.58 ± 0.26 mg leaf^{-1} d^{-1} or 1.1 ± 0.43 % d^{-1} while that of *A. antarctica* (initially 0.64 ± 0.053 g leaf^{-1}) only declined at 2 ± 2.7 mg leaf^{-1} d^{-1} or 0.3 ± 0.41 % d^{-1}. Hence it is surprising that excised *H. ovalis* leaves photosynthesised for so much longer. I observed periphyton colonising *H. ovalis* but not *A. antarctica*. If periphyton contributed to photosynthesis, it may have exacerbated the observed difference in detrital viability between the two seagrasses. Accounting for the uncertainty introduced by these limitations, I expect the average seagrass to have a photosynthetic half-life of more than a week but less than a month.

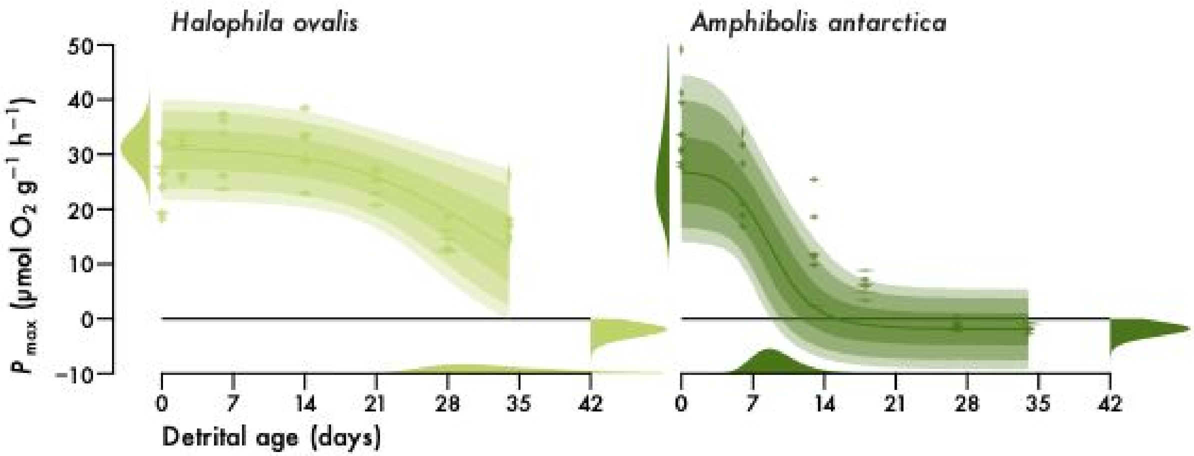


Figure 1. Seagrass detrital photosynthesis. Light-saturated net photosynthesis (P_{max}) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves over 34 days post-excision. P_{max} is standardised by leaf fresh mass. Violins are kernel density estimates of observations with measurement error (n = 42 for *H. ovalis*, n = 39 for *A. antarctica*). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability for predicted observations. Distributions are posteriors for initial P_{max} (α), photosynthetic half-life (μ) and final respiration (τ) correctly mapped onto the axes (cf. Equation 1).

Table 1. Seagrass parameter estimates (mean ± standard deviation). Δ is the absolute difference between estimates for *H. ovalis* (n = 42) and *A. antarctica* (n = 39). $P(\Delta > 0)$ is the probability that estimates for *H. ovalis* and *A. antarctica* are different. P = 0.5 is equivalent to a coin toss.

Parameter	***Halophila ovalis***	***Amphibolis antarctica***	**Δ**	***P*(Δ > 0)**
Mass-based				
α (µmol g^{-1} h^{-1})	31 ± 3.4	28 ± 8.4	3.3 ± 9.1	0.69
μ (days)	35 ± 11	9.5 ± 2.6	26 ± 11	1
τ (µmol g^{-1} h^{-1})	2.1 ± 0.85	1.9 ± 0.76	0.17 ± 1.1	0.56
Leaf-based				
α (µmol $leaf^{-1}$ h^{-1})	1.6 ± 0.17	33 ± 12	31 ± 12	1
μ (days)	46 ± 15	7 ± 1.3	39 ± 15	1
τ (µmol $leaf^{-1}$ h^{-1})	0.92 ± 0.38	0.93 ± 0.37	0.0049 ± 0.53	0.51

Meta-analysis

Comparison of detrital photosynthesis across various plants revealed that light ameliorates photosynthetic decay. Detritus of terrestrial plants, seagrasses and seaweeds with access to light is predicted to have a photosynthetic half-life that is 1.8, 1.6 and 2 times as long as that in darkness (Figure 2, Table 2). Freshwater plants form an exception with a mere 68% chance (only somewhat above 50:50) of extended photosynthesis under light (Figure 2, Table 2). The predicted median light–dark ratio for new plants is therefore 1.5 (ln ratio = 0.4 ± 0.52). In absolute terms, however, the light effect is only substantial for seaweeds (Figure 2). While median terrestrial photosynthetic half-life is only expected to increase from 7.2 to 13 days with access to light, that of seaweeds increases from 24 to 51 days (Table S2). In darkness, seaweed detritus thus only remains viable for comparable durations to other plants (Figure 2, Table S2).

The choice of response variable also matters when estimating detrital photosynthesis. Chlorophyll generally decays slower than photosynthesis. This pattern is consistent across all plants with photosynthetic half-life predicted to be 1.4 times (ln ratio = 0.33 ± 0.36) as long for chlorophyll as for photosynthesis (Figure 2, Table 2). In absolute terms, the effect again naturally dominates for seaweeds (Figure 2), with an increase from 51 days for photosynthesis to 81 days for chlorophyll under light (Table S2). Measurements based on chlorophyll therefore tend to overestimate the longevity of detrital photosynthesis.

Across light conditions and response variables, the median half-life of detrital photosynthesis ranges from 11 days for terrestrial plants, over 14 and 17 days for freshwater plants and seagrasses to 44 days for seaweeds (Table 2). The prediction for new plants is 17 days (ln = 2.8 ± 1.1). In relative terms, seaweed detrital photosynthesis persists 3.9, 3.1, and 2.5 times as long as that of terrestrial plants, freshwater plants and seagrasses (Table 3, Figure S2). All vascular plant detritus remains viable for similar durations but there is a tendency for that of freshwater plants to have a somewhat longer photosynthetic half-life compared to terrestrial plants, with that of seagrasses being longer still (Table 3, Figure S2).

Table 2. Meta-analysis parameter estimates (Equation 2) for plant groups. Photosynthetic half-life (μ) is predicted for new experiments on unobserved species across the light (β) and chlorophyll (γ) effects. Exponentiating β and γ yields relative effect sizes (cf. Figure 2). μ is also

expressed relative to detrital half-life ($t_{½}$) to quantify the physiological influence on decomposition (cf. Figure 3). All parameter estimates are given as median (ln mean ± standard deviation) to accurately represent right-skewed probability distributions. n and P denote sample size and probability.

Parameter	Terrestrial plants	Freshwater plants	Seagrasses	Seaweeds
n	7240	673	160	2573
μ (days)	11 (2.4 ± 0.92)	14 (2.7 ± 0.95)	17 (2.9 ± 0.97)	44 (3.8 ± 0.96)
e^{β}	1.8 (0.61 ± 0.18)	1.2 (0.16 ± 0.37)	1.6 (0.47 ± 0.4)	2 (0.77 ± 0.47)
$P(\beta > 0)$	1	0.68	0.89	0.98
e^{γ}	1.3 (0.29 ± 0.15)	1.5 (0.41 ± 0.26)	1.4 (0.32 ± 0.3)	1.5 (0.47 ± 0.29)
$P(\gamma > 0)$	0.97	0.96	0.89	0.98
$t_{½}$ (days)	409 (6 ± 1.5)	57 (4 ± 1.5)	68 (4.2 ± 1.5)	14 (2.7 ± 1.5)
$\mu/t_{½}$	0.028 (−3.6 ± 1.8)	0.25 (−1.4 ± 1.8)	0.25 (−1.4 ± 1.8)	3 (1.1 ± 1.8)
$P(\mu/t_{½} > 1)$	0.02	0.22	0.23	0.73

Notwithstanding these general trends, there is substantial interspecific and experimental variation in photosynthetic half-life (Figure 3). Median estimates for species range from 3–4 days for the wheat *Triticum aestivum* to 154 days for the Antarctic brown seaweed *Desmarestia anceps* (Table S3). Median estimates for experiments range from 1.2 to 438 days. The variance in photosynthetic half-life between plant groups, species and experiments is directly quantified by logarithmic partial pooling standard deviations (σ_{α} in Equation 2). These standard deviations respectively are 0.51 ± 0.16, 0.53 ± 0.12 and 0.72 ± 0.057 across plant groups, species and experiments, so experimental variation is the greatest source of uncertainty.

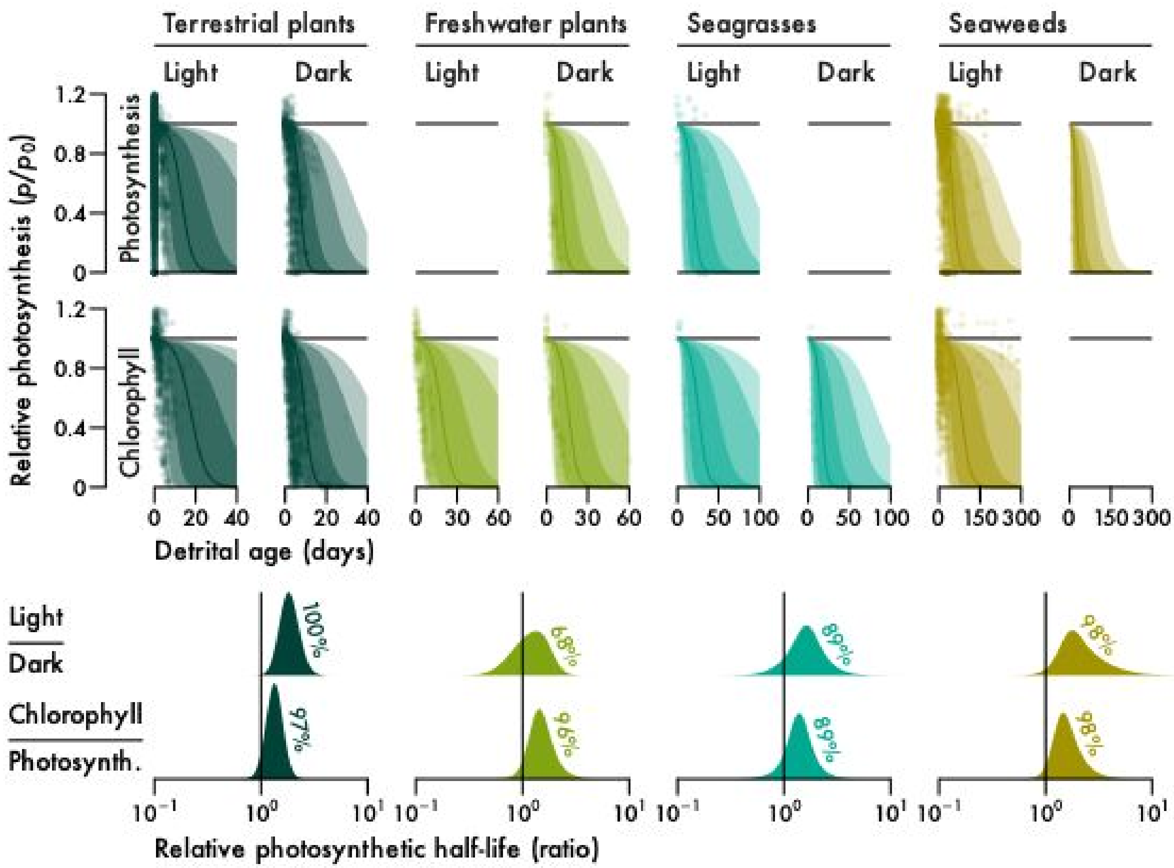


Figure 2. The effect of light and response variable type on detrital photosynthesis for major plant groups: terrestrial plants (Streptophyta excluding aquatic plants), freshwater plants (Alismatales excluding seagrasses), seagrasses, and seaweeds (Chlorophyta and Heterokontophyta excluding freshwater macroalgae). Points are observations (n = 7240, 673, 160 and 2573 for terrestrial plants, freshwater plants, seagrasses and seaweeds). Note that there are several observations outside the axes ranges. Lines and intervals are medians and the central 50, 80 and 90% of posterior probability for the median prediction ($e^{\bar{p}}$ in Equation 2). Note that data are missing where panels are empty and predictions are therefore not shown. Distributions are kernel density estimates of posterior probabilities for the light and chlorophyll relative effect sizes (e^{β} and e^{γ} in Equation 2). Percentages are probabilities for $\beta > 0$ and $\gamma > 0$.

Decomposition timescales also vary substantially across plant groups. Terrestrial plant detritus decomposes very slowly, only reaching half its initial mass after over a year on average (Figure 3, Table 2). Aquatic angiosperms decompose much faster, with a median detrital half-life ranging from 57 days for freshwater plants to 68 days for seagrasses (Figure 3, Table 2), only 14% to 17% of terrestrial half-life (Table 3). Seaweeds decompose fastest, with a median detrital half-life of only two weeks (Figure 3, Table 2), 21–26% of aquatic angiosperm half-life and only 3.5% of terrestrial half-life (Table 3). Photosynthetic half-life must therefore be viewed in relation to detrital half-life to assess how affected decomposition is by detrital photosynthesis.

The ratio of photosynthetic to detrital half-life ($\mu/t_{½}$) quantifies the relative influence of detrital photosynthesis on decomposition, where ratios below 1 indicate little influence and ratios of 1 and above indicate substantial influence. The median of $\mu/t_{½}$ ranges from 0.028 for terrestrial plants over 0.25 for freshwater plants and seagrasses to 3 for seaweeds (Figure 3, Table 2). Seaweeds are therefore the only plants with a median ratio above 1 and there is a 73% chance that the ratio for any seaweed is above 1, while the probability for aquatic angiosperms is 22–23% and only 2% for terrestrial plants (Figure 3, Table 2).

Table 3. Photosynthetic (μ) and detrital ($t_{½}$) half-life contrasts between plant groups (cf. Figure S2). P(ratio > 1) is the probability that estimates are different. P = 0.5 is equivalent to a coin toss. All estimates except P are given as median (ln mean ± standard deviation). See Table 2 for sample sizes.

Ratio	μ/μ	$P(\mu/\mu > 1)$	$t_{½}/t_{½}$	$P(t_{½}/t_{½} > 1)$
Seaweeds / Terrestrial	3.9 (1.4 ± 1.3)	0.85	0.035 (−3.3 ± 2.1)	0.057
Seaweeds / Freshwater	3.1 (1.1 ± 1.4)	0.80	0.26 (−1.4 ± 2.1)	0.26
Seaweeds / Seagrasses	2.5 (0.93 ± 1.4)	0.75	0.21 (−1.6 ± 2.1)	0.23
Seagrasses / Terrestrial	1.5 (0.43 ± 1.3)	0.63	0.17 (−1.8 ± 2.1)	0.2
Freshwater / Terrestrial	1.3 (0.23 ± 1.3)	0.57	0.14 (−2 ± 2.1)	0.18
Seagrasses / Freshwater	1.2 (0.2 ± 1.3)	0.56	1.2 (0.18 ± 2.2)	0.53

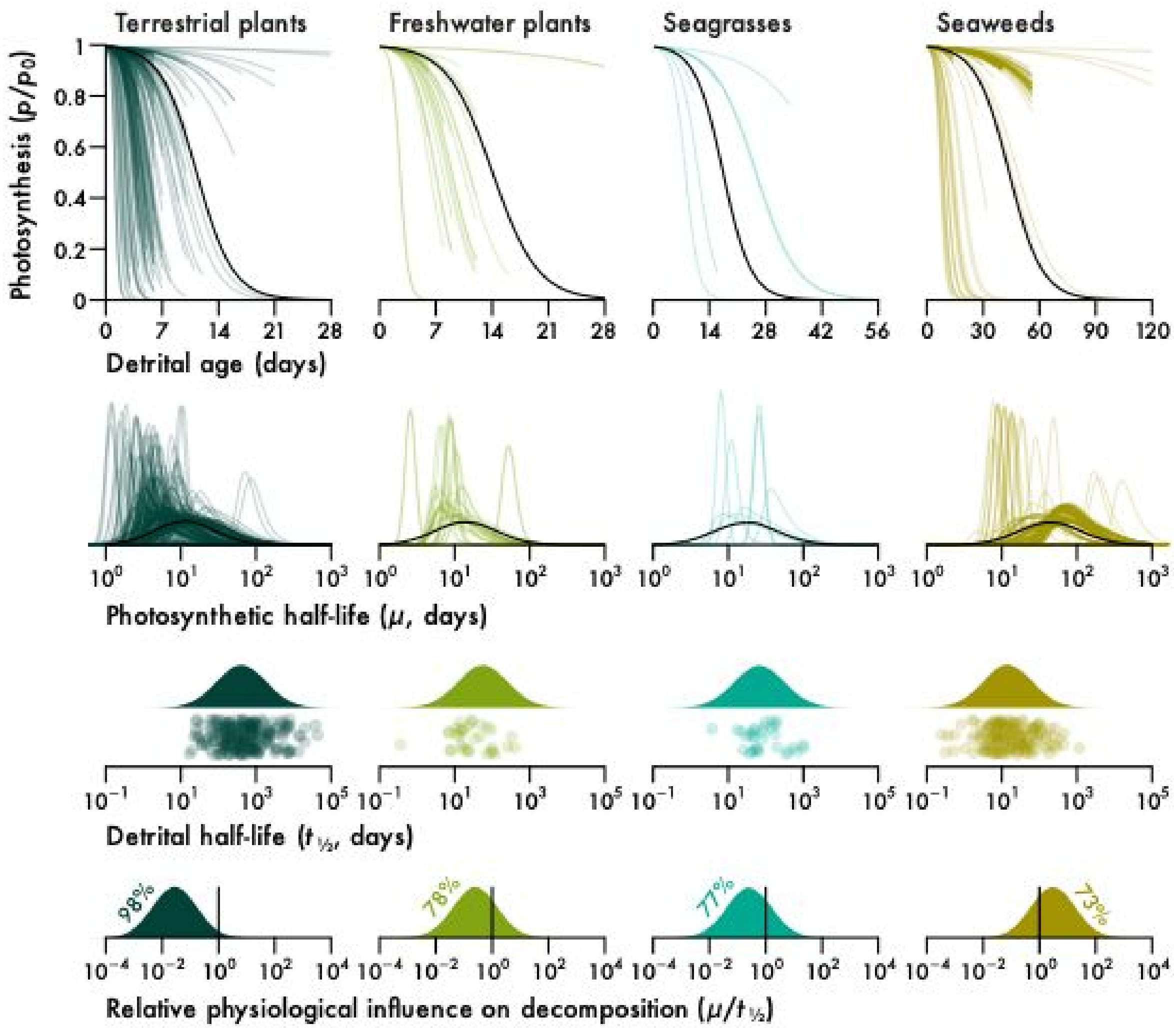


Figure 3. Longevity of detrital photosynthesis relative to detrital mass for major plant groups (cf. Figure 2). Lines are median predictions ($e^{\bar{p}}$ in Equation 2) or posterior probability distributions of photosynthetic half-life (μ in Equation 2). Existing experiments (n = 314, 32, 8 and 197 for terrestrial plants, freshwater plants, seagrasses and seaweeds) are given in colour and predictions for new experiments on unobserved species across the light and chlorophyll effects are given in black. Raincloud plots of detrital half-life ($t_{½}$ sensu Olson 1963) show observations (n = 239, 33, 32 and 171 for terrestrial plants, freshwater plants, seagrasses and seaweeds) below posterior probability distributions for new observations. Distributions in the bottom panel are ratios of the black distributions of μ to the distributions of $t_{½}$. Percentages are probabilities for $\mu/t_{½} < 1$ or $\mu/t_{½} > 1$, depending on what side of the identity line most of the probability lies.

Discussion

Some plant detritus can maintain photosynthesis for long periods following detachment. This is at odds with conventional understanding of detritus as dead organic matter. Despite the existence of various relevant literature published on this topic since the 1950s, the phenomenon of detrital photosynthesis had not been explored beyond kelps. In part, this is probably due to diverse research objectives and terminology which obscure the connection between quantitatively comparable data. There was also data scarcity for seagrasses which, as the only vascular counterpart to seaweeds, form an important functional link to freshwater and terrestrial plants. Here I collected the first data on photosynthesis in seagrass detritus, developed a mathematical model of detrital photosynthesis and used it to make the first quantitative comparison of photosynthetic persistence in the detrital phase of various plants. Results suggest that the non-vascular macroalgae have the greatest potential for detrital photosynthesis while among the vascular plants there is slightly increased detrital viability with a shift from a terrestrial to an aquatic and yet again to a marine lifestyle. Although all plants display detrital photosynthesis to some extent, it does not necessarily influence their decomposition. Terrestrial plant decomposition is slower than that of aquatic vascular plants which in turn is much slower than that of macroalgae. I predict that detrital photosynthesis is a general phenomenon but its effect on decomposition ranges from negligible for terrestrial plants over measurable for aquatic vascular plants to substantial for macroalgae. It is possible that detrital photosynthesis determines macroalgal decomposition. I therefore propose a new causal model of macroalgal decomposition that emerges from this formalisation of the theory of detrital photosynthesis (Figure 4). While conventional decomposition theory may continue to be applied to vascular plants and detritus sensu stricto, its assumptions will not always hold for aquatic systems and certainly do not apply to macroalgae (Figure 4).

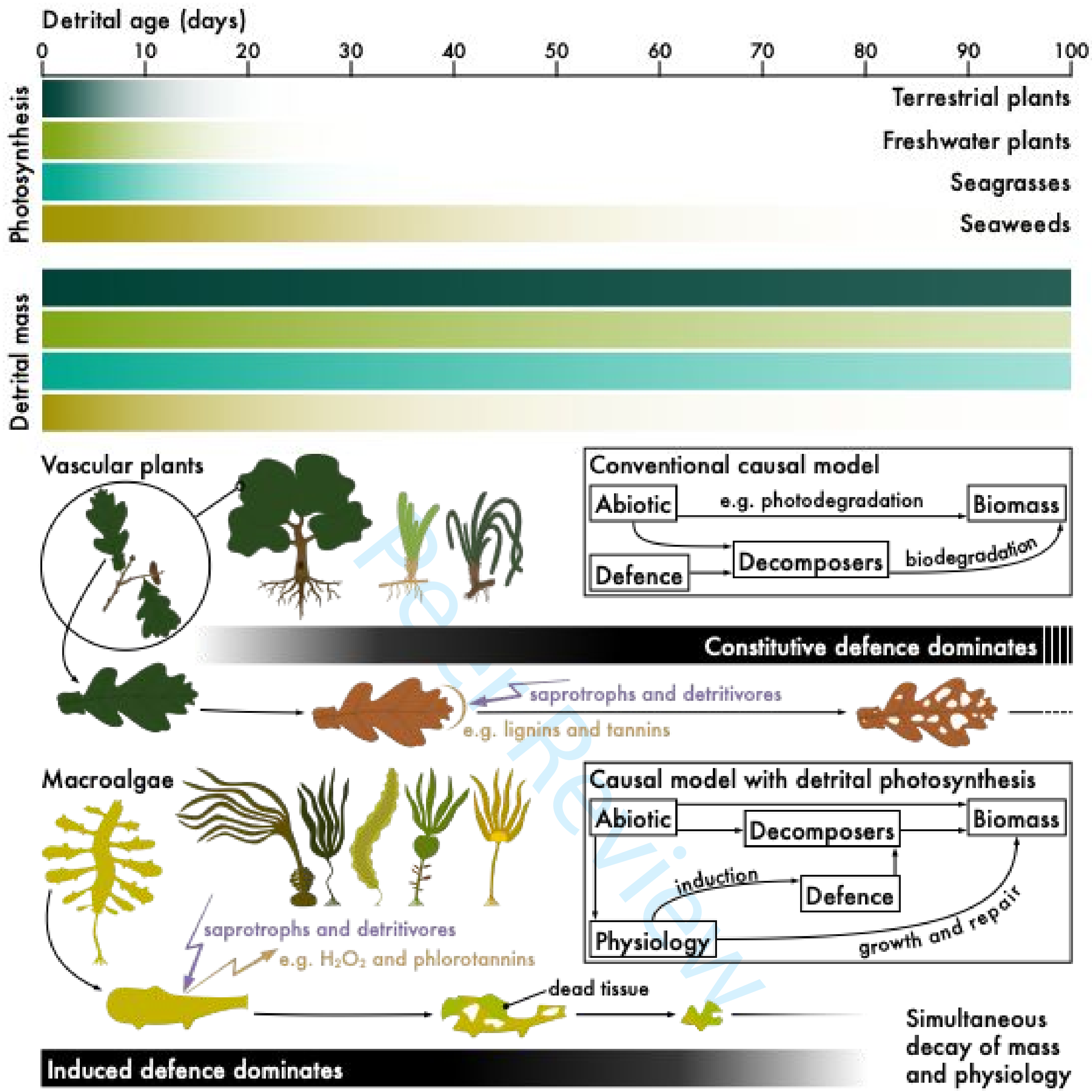


Figure 4. Conceptual model of decomposition in vascular plants and macroalgae. Coloured bars show the decay of detrital photosynthesis and biomass for major plant groups based on the median of p (Equation 2) and e^{-kt} where k is the median of ln2 / $t_{½}$ (cf. Figure 3). Black bars show approximate timelines for the dominance of constitutive and induced defences in vascular plants and macroalgae respectively. Boxes show the conventional and new causal models as directed acyclic graphs where nodes are variables and arrows show their causal relationships. Some arrows are labelled with mechanisms for clarity.

Detrital photosynthesis per se is not restricted to non-vascular plants. Seagrass detritus can also photosynthesise for at least a week. The oxygen production of the diverse seagrasses *Halophila ovalis* and *Amphibolis antarctica* responds similarly to detrital age and the delayed response in *H. ovalis* may simply be attributable to greater colonisation by periphyton. Given a leaf mass per area of 2.7 and 6.8 mg cm^{-2} for *H. ovalis* and *A. antarctica* respectively (de los Santos et al. 2012, 2016), periphyton could contribute 12–172% of initial seagrass photosynthesis (Neely & Wetzel 1997). Alternatively, *A. antarctica* leaves may be more reliant on their roots for nutrient provisioning, although there is currently no evidence to suggest this (Pedersen et al. 1997). The 26–39-day shorter photosynthetic half-life of isolated *A. antarctica* leaves compared to *H. ovalis* is surprising given their 1.7 times or 47-day longer attached lifespan (Hemminga et al. 1999). Nonetheless, recent evidence suggests that detached *H. ovalis* can continue growing to a similar or greater extent relative to other seagrasses (Hanuise et al. 2026). Despite such observations of apparent detrital growth for up to a month (Hanuise et al. 2026) and my evidence of photosynthetic persistence for a similar duration, seagrass detrital photosynthesis may not necessarily counteract decomposition. I found that seagrass detritus has a detrital half-life of 68 days on average and complete decomposition may take a year (Harrison 1989, Cebrián et al. 1997, Cebrián & Lartigue 2004, Trevathan-Tackett et al. 2020). This is much longer than detached seagrass is expected to photosynthesise and accordingly there is little evidence of physiological influence on its decomposition. While live *Halophila decipiens* ramets do indeed decompose slower than buried ones (Kenworthy et al. 1989), decomposition rates are similar between live and senescent or buried *Thalassia testudinum* leaves (Rublee & Roman 1982, Fourqurean & Schrlau 2003, Rosch & Koch 2009) and reviews suggest that live seagrass leaves actually tend to decompose faster than senescent or dead ones (Harrison 1989, Trevathan-Tackett et al. 2020).

An aquatic lifestyle is conducive to detrital photosynthesis but secondary to non-vascularity. Macroalgae uniquely exhibit long-term detrital photosynthesis which clearly sets them apart from vascular plants (Figure 4). Nevertheless, seagrasses and freshwater plants on average display detrital photosynthesis several days longer than terrestrial plants. This is almost certainly due to the greater photosynthetic emancipation of aquatic plant leaves. Because most experiments are carried out in water, they probably overestimate the persistence of detrital photosynthesis in terrestrial plants. With access to light but not water, which is the normal fate of terrestrial plant detritus, detached leaves of various trees tend to lose the ability to photosynthesise in just a few minutes (e.g. Gauthier & Jacobs 2018, Kar et al. 2021) due to breakdown of stomatal conductance following a loss in water pressure (Powles et al. 2006, Gauthier & Jacobs 2018). The true difference between aquatic and terrestrial vascular plants may therefore be greater. Surprisingly, seagrasses detritus tends to photosynthesise for longer than that of freshwater plants from the same order. The adoption of a marine lifestyle and convergent evolution with seaweeds (Olsen et al. 2016) may be the ultimate reason. But a more proximate explanation is excess production of hydrogen peroxide in detached freshwater plant leaves (Jana & Choudhuri 1982, Kar & Choudhuri 1986, 1987). As demonstrated for *Hydrilla verticillata* (Kar & Choudhuri 1986, 1987), hydrogen peroxide production is exacerbated by photosynthesis, leading to light-induced chlorophyll degradation that is independent of photodegradation. Hence freshwater plants were the only group for which I did not find a positive light effect on photosynthetic half-life. In contrast, there is no evidence for increased chlorophyll breakdown in isolated seagrass leaves exposed to light (Strother & Vatta 1986) and seagrass photodegradation has only been reported for dead leaves (Vähätalo et al. 1998). Perhaps the answer lies in alterations to the light harvesting complexes and cell wall features of seagrass leaves that enhance gas exchange (Olsen et al. 2016). Regardless, emancipation of leaf photosynthesis from the rest of the plant is apparently favoured by a marine even more so than an aquatic lifestyle.

Macroalgal decomposition is uniquely determined by detrital photosynthesis. Of all considered plants, terrestrial ones decompose slowest and macroalgae decompose fastest, so the observed divergence in photosynthetic decay is increased when viewing it relative to decomposition. Macroalgae are the only plants whose detritus may photosynthesise for as long as it remains intact, aquatic vascular plants maintain photosynthesis for a quarter of their detrital phase and terrestrial decomposition is probably unaffected by physiology due to a mismatch between rapid photosynthetic decay and slow decomposition (Figure 4). A clear distinction must here be made between vascular plants and macroalgae. Since the majority of vascular plant decomposition occurs in the absence of physiology, its main intrinsic predictor must be constitutive defence while that of macroalgal decomposition is likely induced defence (Figure 4). Increases in phlorotannin content in kelp detritus (Tala et al. 2019, de Bettignies et al. 2020, Wright et al.

2022) provide some evidence for this but other induced defences such as hydrogen peroxide release have yet to be measured in detritus. Detrital photosynthesis can further affect detrital mass through growth and repair and introduces novel causal paths (Figure 4). For example, light prolongs detrital photosynthesis which may slow macroalgal decomposition through growth and provide a mechanism contrary to photodegradation. In this case the result determines the mechanism, but the same result can also be produced by different mechanisms. Elevated temperature may accelerate decomposition in three ways: by reducing the structural integrity of detritus, by increasing decomposer metabolism, and also by diminishing detrital photosynthesis as was recently demonstrated (Wright et al. 2024). Further research is required to tease apart these causal relationships.

Detrital photosynthesis must be considered when making inferences about aquatic plant decomposition. The initial physiological state of detritus and subsequent physiological stressors in the experiment may not reflect those in nature. In contrast to seagrass litter bag experiments (Harrison 1989, Trevathan-Tackett et al. 2020), most seaweed decomposition experiments have used freshly excised blades as a detritus proxy (Frontier et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Wright et al. 2022, Kennedy & Blain 2025, White & Norkko 2025). Senescent or naturally detached fronds are rarely used (Hamersley et al. 2015, de Bettignies et al. 2020) and the practice of pre-killing tissue to obtain detritus sensu stricto is restricted to few, mostly early studies (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). While the fraction of seaweed detritus that consists of whole dislodged plants or fronds is certainly initially viable, the majority of tissue that is abscised or eroded likely exhibits impaired photosynthesis from the start. Ignorance of the causation introduced by the presented theory (Figure 4) would lead to the erroneous conclusion that decomposition rates measured on excised tissue are representative. Detrital photosynthesis is highly sensitive to light in most plants and may also be sensitive to temperature (Wright et al. 2024) and more nuanced environmental factors that remain overlooked. Awareness of the presented theory is necessary to control for these variables. For example, decomposition rates measured under any influence of even weak light will be highly misleading when the goal is to predict carbon cycling in the deep ocean (Filbee-Dexter et al. 2024, White & Norkko 2025). This is not immediately obvious without the causal link provided by detrital photosynthesis (Figure

4). Detrital photosynthesis likely accounts for the large unexplained variation in macroalgal decomposition rates observed across experiments (Kennedy & Blain 2025, White & Norkko 2025).

To conclude, I show that all plants may photosynthesise following detachment, but timescales vary greatly from a few minutes in tree leaves without access to water to a year for certain seaweeds. Only macroalgal detrital photosynthesis decays on similar timescales to detrital mass. Physiology is therefore expected to influence the decomposition of macroalgae, a causal relationship which is unique among plants. The initial physiological state of detritus and predictors of photosynthesis, such as light and temperature, can hence be used to predict macroalgal decomposition. With the theory of detrital photosynthesis formalised, we now need (1) a mathematical model of macroalgal decomposition that takes this theory into account — I have recently made some progress in this direction (Wright 2026b), (2) further studies on post-excision photosynthesis and induced defence, especially in underrepresented plants such as seagrasses and seaweeds other than kelps, (3) decomposition experiments comparing live and dead detritus to directly investigate the proposed effect of physiology on decomposition (cf. Birch et al. 1983, Brouwer 1996), (4) studies on the initial physiological state of detritus produced by erosion, abscission and dislodgement, and (5) in situ measurements of detrital photosynthesis in detrital accumulations.

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Conflicts of interest

I declare no conflicts of interest.

Data and code availability

Data and annotated code are available at github.com/lukaseamus/plant-limbo. The dataset used for meta-analysis (Wright 2025) can also be accessed at github.com/lukaseamus/detrital-photosynthesis.

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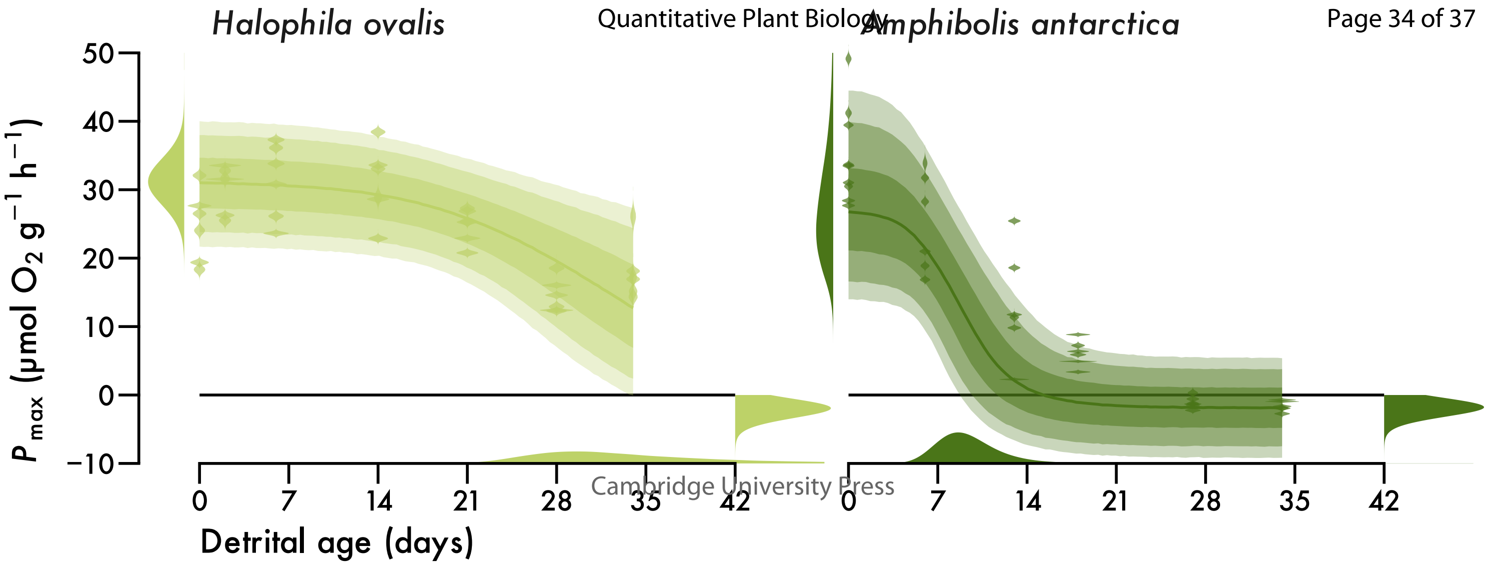
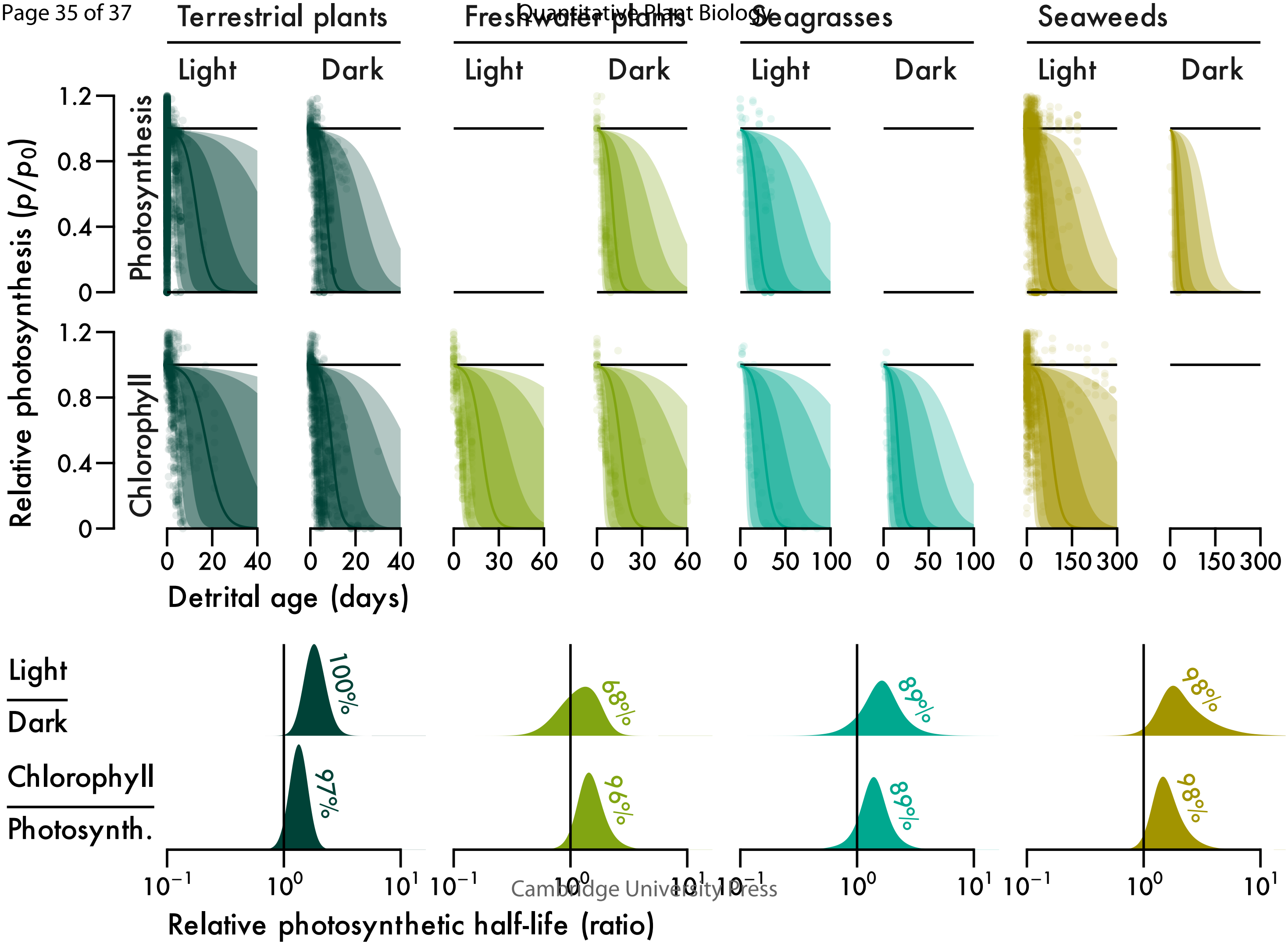
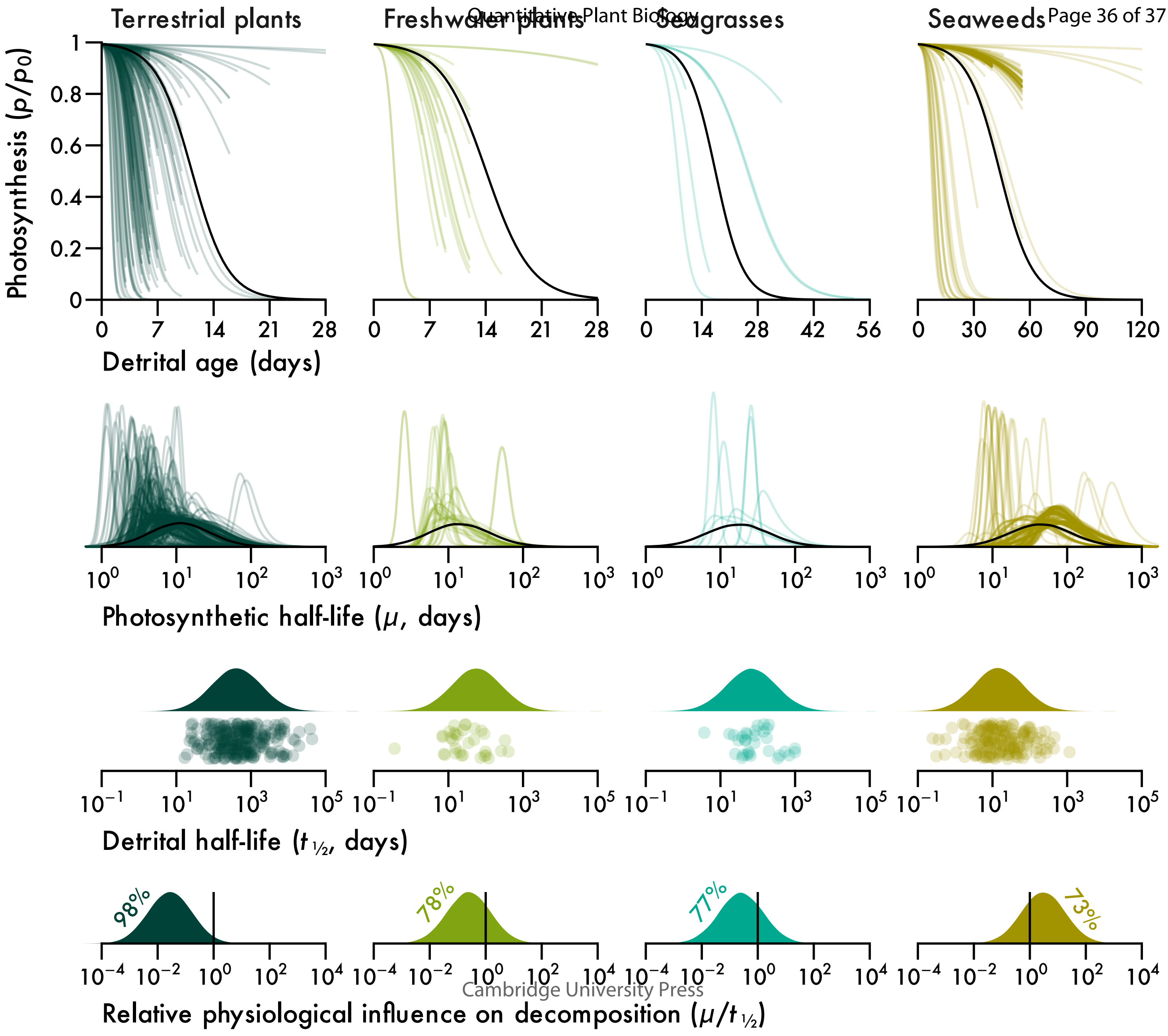
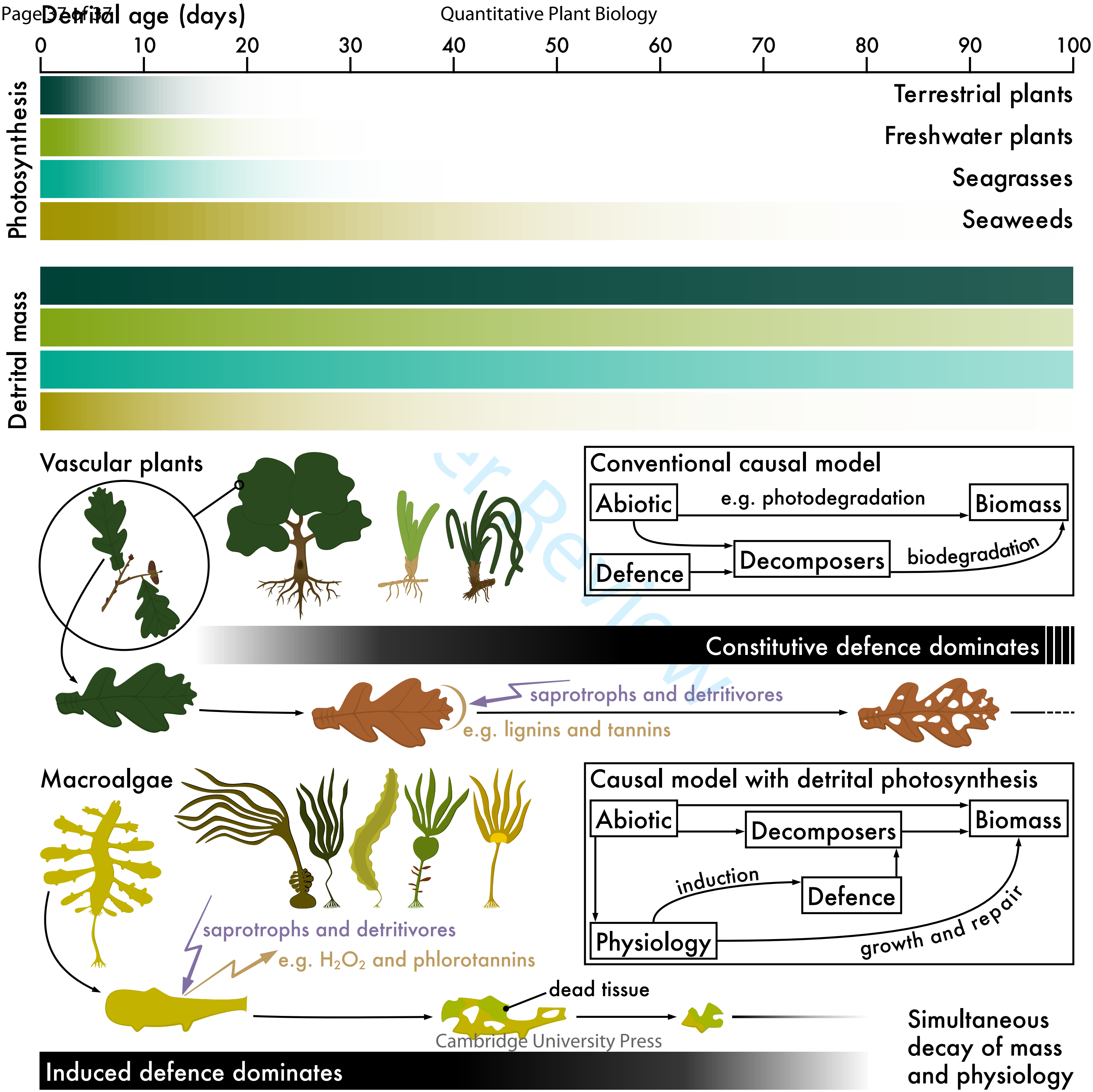

Halophila ovalis
Amphibolis antarctica
50
40
30
20
10
0
−10
P_{max} (μmol O_2 g^{-1} h^{-1})
0
7
14
21
28
35
42
0
7
14
21
28
35
42
Detrital age (days)


Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Light
Dark
Photosynthesis
Chlorophyll
Relative photosynthesis (p/p_0)
0
0.4
0.8
1.2
0
20
40
30
60
50
100
150
300
Detrital age (days)
Light
Dark
Chlorophyll
Photosynth.
100%
89%
89%
98%
97%
96%
89%
98%
10^{-1}
10^{0}
10^{1}
Relative photosynthetic half-life (ratio)



Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Photosynthesis (p/p_0)
1
0.8
0.6
0.4
0.2
0
0 7 14 21 28
0 7 14 21 28
0 14 28 42 56
0 30 60 90 120
Detrital age (days)
10^0 10^1 10^2 10^3
Photosynthetic half-life (μ, days)
10^{-1} 10^1 10^3 10^5
Detrital half-life ($t_{1/2}$, days)
98%
78%
77%
73%
10^{-4} 10^{-2} 10^0 10^2 10^4
Relative physiological influence on decomposition ($\mu/t_{1/2}$)


Detrital age (days)
0
10
20
30
40
50
60
70
80
90
100
Photosynthesis
Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
Detrital mass
Vascular plants
Conventional causal model
Abiotic
e.g. photodegradation
Biomass
Defence
Decomposers
biodegradation
Constitutive defence dominates
saprotrophs and detritivores
e.g. lignins and tannins
Macroalgae
Causal model with detrital photosynthesis
Abiotic
Decomposers
Biomass
induction
Defence
Physiology
growth and repair
saprotrophs and detritivores
e.g. H_2O_2 and phlorotannins
dead tissue
Induced defence dominates
Simultaneous decay of mass and physiology